

Homothetic Toroidal Models

Geometry, Phase Reconstruction, and Display Information

Hilmir Frímann Halldórsson

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Abstract

A toroidal formula can specify a surface, an observation on a parameter domain, or a display of recorded data. We study these meanings separately for a finite homothetic family of toroidal charts and a prescribed Gaussian-weighted phase sum. We obtain an exact criterion for intersection of the full surface images, and distinguish it from strict containment of associated diagnostic spindle bodies. The squared modulus of the phase sum depends on the prescribed phase through a span of dimension at most three, $\text{span}\{1, \cos(2\Phi), \sin(2\Phi)\}$, and is recovered from three known phase samples under a sharp nonsingularity condition. A reflection and exponential-independence argument excludes nonzero rigid azimuthal rotation under explicit ring-family hypotheses. We derive the induced area measures, identify their normalization rules, and exhibit nonunique ambient extensions of the reference-shell observation. Finally, exact inverses and exceptional fibres characterize three separate display maps, including the difference between coordinates and a richer retained record. The results are mathematical consequences of stated definitions; they do not supply an evolution law or a physical interpretation. Coordinate-based figures and isolated algebraic and numerical checks accompany the written proofs.

1 Question, definitions, and scope

What geometry and information are determined by a scaled toroidal construction? A useful answer requires distinguishing the domain of a formula from its image, and an observation from a display of that observation. This paper develops that distinction for one finite family whose radii form geometric progressions. It then analyzes the phase dependence of a scalar observation and the information retained by separate coordinate displays. Each construction is defined here, so the arguments do not require access to historical documents or software.

The geometric starting point is classical: a circle rotated about an axis produces ring, horn, or spindle images according to the ratio of its major and minor radii [15]. Circle-intersection algebra is also classical [14]. The contribution of Sections 2–3 is their application to a fixed-aspect homothetic family, with both signed meridian branches retained. Sections 4–5 give a finite phase reduction and a restricted rotation obstruction. Three-sample recovery is an elementary instance of trigonometric interpolation; it is related in method, but not identical in sampling protocol, to phase-shifting reconstruction [1]. Sections 6–7 concern measures, extension freedom, and exact inverses of explicitly defined maps. We make no claim of priority for the underlying geometry or interpolation principles.

The historical motivation is the common-angle toroidal sum in *Converging Symmetries* [9, 8] and its later presentation in *Recursive Field Collapse into Toroidal Attractor Structures* [7]. These sources and separate display code motivated the questions, but do not identify all of their objects with one another. The short source correspondence in Appendix A records

the limited connection. The present exposition incorporates the final reconciled reference and subsequent accepted qualifications [5, 2, 4].

Throughout, $N \geq 1$ is a finite integer, $s > 1$, and $r_0, R_* > 0$. Indices satisfy $0 \leq n < N$. Angles belong to $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$. Define

$$r_n = r_0 s^n, \quad R_n = R_* s^{n-N/2}, \quad \bar{R} = R_* s^{-N/2} = R_{n=0}, \quad a = \frac{\bar{R}}{r_0}. \quad (1)$$

The letters R_n and r_n denote geometric radii; C below is an independent Gaussian-centre parameter. No equality between C and R_* is assumed. Spatial coordinates are Euclidean. If a length unit is assigned, all radii and C have that unit and Gaussian precision α has inverse-square length units. The source does not determine a physical unit for the complex amplitude or its squared modulus. Angles and their trigonometric arguments are dimensionless.

The central distinctions can be stated before the proofs. A chart $X_n : \mathbb{T}^2 \rightarrow \mathbb{R}^3$ has a full image S_n . A body K_n , introduced later, is a separate diagnostic construction. An observation ψ is initially a function on the common angle domain; its nonnegative squared modulus P becomes a normalized density only after a measure and a positive integral are specified. A display map acts on other recorded quantities and may discard information. Reusing a toroidal coordinate expression does not provide a mathematical identification between these objects.

2 Carriers: homothety and singular charts

Define

$$X_n(\theta, \phi) = ((R_n + r_n \cos \theta) \cos \phi, (R_n + r_n \cos \theta) \sin \phi, r_n \sin \theta), \quad S_n = X_n(\mathbb{T}^2). \quad (2)$$

The abstract torus $\mathbb{T}^2$ is always a smooth parameter domain. Whether (2) is an embedding is a different question.

Proposition 2.1 (Common aspect and chart classification). *Let $Y = X_0$ with major radius $\bar{R}$ and minor radius r_0 . Then $X_n = s^n Y$ and $R_n/r_n = a$ for every admissible n . If $a > 1$, each chart is an embedding and S_n is a ring torus. For $a = 1$ the chart is noninjective and loses rank along $\theta = \pi$. For $0 < a < 1$ it is noninjective and loses rank along the two parameter circles $\cos \theta = -a$.*

Proof. Substitution of (1) into each component of (2) factors out s^n . Write $q = R_n + r_n \cos \theta$. Direct differentiation gives

$$|X_{n,\theta}|^2 = r_n^2, \quad X_{n,\theta} \cdot X_{n,\phi} = 0, \quad |X_{n,\phi}|^2 = q^2. \quad (3)$$

For a ring, $q > 0$. The spatial cylindrical radius $\rho = \sqrt{x^2 + y^2}$ equals q , and the point determines $\phi = \text{atan2}(y, x)$ and $\theta = \text{atan2}(z, \rho - R_n)$ modulo 2π . Thus the chart is injective and has rank two. A continuous injection from the compact domain into the Hausdorff space $\mathbb{R}^3$ is a homeomorphism onto its image, proving that this immersion is an embedding.

For a horn, $q = 0$ exactly at $\theta = \pi$. Every value of ϕ there maps to the origin, and $X_{n,\phi} = 0$ while $|X_{n,\theta}| = r_n > 0$. For a spindle, $q = 0$ at two values of θ , with respective heights $z = \pm r_n \sqrt{1 - a^2}$. Each entire parameter circle collapses to one of these axis points, and the rank is again one there. These collapses establish noninjectivity without requiring a classification of every fibre elsewhere. $\square$

The family has one aspect ratio, not a succession of different ring/horn/spindle types. Holding R_*/r_0 and s fixed, the equation for a horn is

$$N = \frac{2 \log(R_*/r_0)}{\log s}. \quad (4)$$

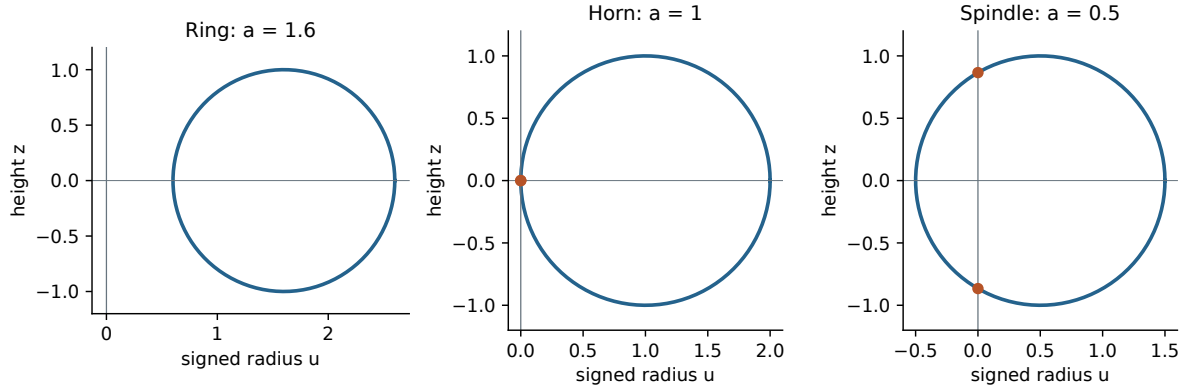


Figure 1: Signed meridian charts $(u, z) = (a + \cos \theta, \sin \theta)$ at minor radius one. The vertical line is the axis of revolution. Ring, horn, and spindle cases use the declared illustrative aspects $a = 1.6, 1, 0.5$. Rotation uses the signed coordinate u ; negative u is not to be discarded when forming the full image.

It determines an admissible model only if the right side is a positive integer. It need not be one. Statements comparing two shells also require $N \geq 2$ and valid indices; an arbitrary real solution of (4) supplies neither condition.

For fixed common angles, every X_n lies on the ray through Y . Two different indices give the same point at these angles only when $Y = 0$, which occurs precisely at the horn collapse. This observation does *not* decide whether the full images intersect: two charts can reach the same spatial point at different angles. The next section treats that question.

3 Image intersection and a separate body diagnostic

3.1 The full-image criterion

Fix admissible indices $n, n + k$ with $k \geq 1$ and $n + k < N$. Put $r = r_n$ and $h = s^k > 1$. The second shell has radii (ahr, hr) while the first has radii (ar, r) .

Theorem 3.1 (Intersection of full images). *The two surface images meet if and only if*

$$L(h) := \frac{h-1}{h+1} \leq a \leq U(h) := \frac{h+1}{h-1}. \quad (5)$$

The endpoints are tangencies of the appropriate meridian circles. For $a = 1$, the two images meet only at the origin.

Proof. In a signed meridian plane use coordinates (u, z) and the circle

$$(u - ar)^2 + z^2 = r^2.$$

Rotation about the z axis maps u and $-u$ to the same cylindrical radius. Consequently a spatial point can be on both images either from two same-side meridian circles or from one circle and the reflection of the other. This covers every case: for any common point, each signed radius is one of $\rho, -\rho$, where $\rho = \sqrt{x^2 + y^2}$; a common reflection makes the first sign positive or negative as needed, leaving only those two comparisons.

For completeness, two circles of radii r and hr , whose centres are separated by $d > 0$, meet precisely when $(h-1)r \leq d \leq (h+1)r$. To see sufficiency as well as necessity, place their

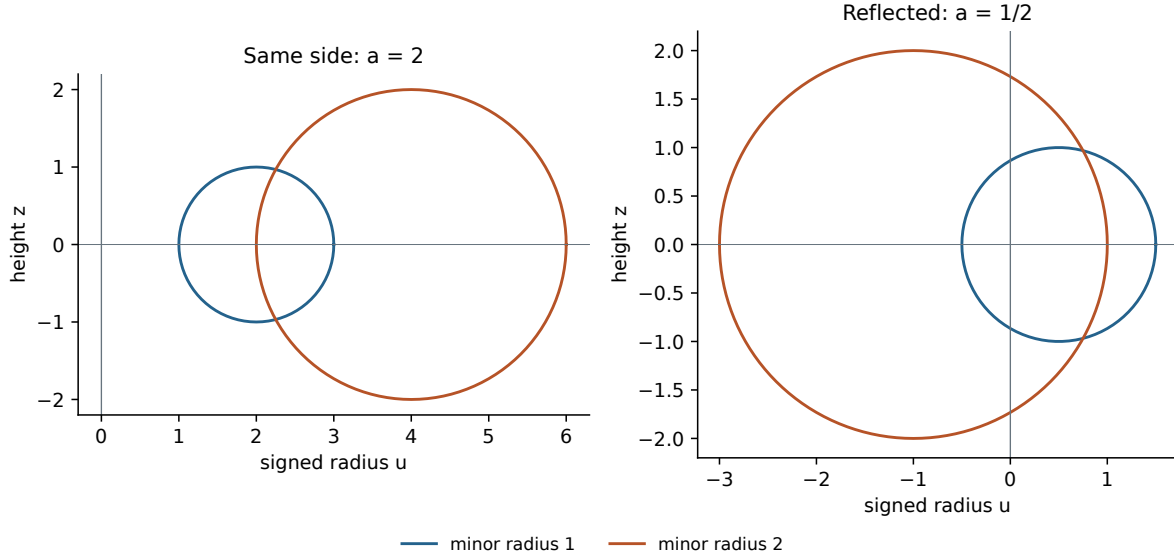


Figure 2: Meridian comparisons at $r = 1$, $h = 2$. Left: $a = 2$ uses same-side circles. Right: $a = 1/2$ uses the reflection of the larger circle. Both lie in the exact interval $[1/3, 3]$. The reflected signed coordinate is a geometric device for testing full spatial images, not an additional shell.

centres at $(0, 0)$ and $(d, 0)$. Subtracting the circle equations gives

$$u_* = \frac{d^2 + r^2 - h^2 r^2}{2d}, \quad z_*^2 = r^2 - u_*^2 = \frac{((h+1)^2 r^2 - d^2)(d^2 - (h-1)^2 r^2)}{4d^2}.$$

The last expression is nonnegative exactly on the stated interval; then $(u_*, \pm\sqrt{z_*^2})$ supplies the intersection. The same inequalities are the triangle inequalities for a common point. Equality gives tangency. This is the usual circle calculation [14].

The same-side centres have separation $d = a(h-1)r$, so this case occurs exactly for $1 \leq a \leq U(h)$. Reflecting the second circle gives separation $d = a(h+1)r$, hence $L(h) \leq a \leq 1$. Their union is (5). Rotating a meridian intersection gives a spatial one, so neither branch is merely a necessary condition.

At $a = 1$ the same-side circles are internally tangent at the origin; the reflected pair is externally tangent there. Thus the origin is the only common spatial point. At $a = L(h)$ the reflected circles are internally tangent, and at $a = U(h)$ the same-side circles are externally tangent. These latter tangent points have nonzero cylindrical radius, so each rotates into a common circle in three dimensions. $\square$

In particular, ignoring the reflected circle removes the entire spindle portion $L(h) \leq a < 1$. Counting same-angle coincidences removes almost all intersections instead. The theorem concerns sets S_n , including their singular images, and makes no statement about an overlap volume or a dynamical interaction.

3.2 Strict containment of diagnostic spindle bodies

Define the closed set

$$K_n = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2} - R_n)^2 + z^2 \leq r_n^2\}. \quad (6)$$

This is a chosen diagnostic body. We do not adopt it as a recovered material region or attach a physical density to its volume.

Theorem 3.2 (Containment despite intersecting images). *If $0 < a < 1$ and $m > n$, then $K_n \subset \text{int } K_m$. Writing $\Delta r = r_m - r_n$, the defining radial distance has a strict margin at least $(1 - a)\Delta r > 0$. Furthermore $S_n \subset K_n$, but the spindle portion with $q = R_n + r_n \cos \theta < 0$ lies in the interior of K_n rather than on its boundary.*

Proof. For any point let $v_n = (\rho - ar_n, z)$, where $\rho = \sqrt{x^2 + y^2} \geq 0$. If the point is in K_n , the triangle inequality gives

$$|v_m| \leq |v_n| + a\Delta r \leq r_n + a\Delta r = r_m - (1 - a)\Delta r < r_m.$$

Continuity of the defining distance makes this point interior to K_m . This proof applies to every point in the body, including axis points.

On the chart, $\rho = |q|$ and $z = r_n \sin \theta$. If $q \geq 0$, $(\rho - R_n)^2 + z^2 = r_n^2$. If $q < 0$, expansion instead gives

$$(\rho - R_n)^2 + z^2 = r_n^2 + 4R_n q < r_n^2. \quad (7)$$

Thus all of S_n is in K_n , but its negative- q sheet is interior. The positive- q sheet, with the seams $q = 0$, is exactly the boundary of K_n : equality in (6) reconstructs $\cos \theta = (\rho - R_n)/r_n$, $\sin \theta = z/r_n$ and $q = \rho$. At the spindle axis seams, arbitrarily small vertical changes reach both sides of the defining inequality, so they too are boundary points. $\square$

Here is an exact witness that makes the distinction concrete. Take $a = 1/2$ and minor radii 1, 2. Then

$$p = \left(\frac{3}{4}, 0, \frac{\sqrt{15}}{4} \right) \quad (8)$$

is on both images. For the small chart choose $\cos \theta = 1/4$, $\sin \theta = \sqrt{15}/4$, $\phi = 0$. For the large chart choose $\cos \theta = -7/8$, $\sin \theta = \sqrt{15}/8$, $\phi = \pi$. The large chart has $q = -3/4$, hence its spatial x is $3/4$. The squared distance in (6) for the larger body is $(3/4 - 1)^2 + 15/16 = 1 < 4$. The point is therefore on the smaller boundary, on the larger inner image sheet, and strictly inside the larger body.

The containment proof does not use $L(h)$, and remains valid below that image-intersection threshold. At $a = 1$ the same inequality only gives nonstrict containment and the common origin is a boundary point. The strict spindle margin, not the lower intersection threshold, is what establishes the theorem.

3.3 A comparison with fixed aspect

For adjacent shells with $s = \varphi_g = (1 + \sqrt{5})/2$,

$$L(\varphi_g) = \varphi_g^{-3}, \quad U(\varphi_g) = \varphi_g^3.$$

As $h > 1$ increases, $L(h)$ increases and $U(h)$ decreases. Thus, among adjacent scale choices $\varphi_g < e < \pi$, the golden-ratio choice has the widest interval of intersecting *aspects*. For example, at fixed $a = 3$ and $N \geq 2$, adjacent golden-ratio shells meet, whereas the choices e and π do not. This compares the criterion, not intersection volume, constructive interference, or stability. Holding R_*/r_0 and N fixed instead changes $a = (R_*/r_0)s^{-N/2}$ when s changes, so it is a different comparison.

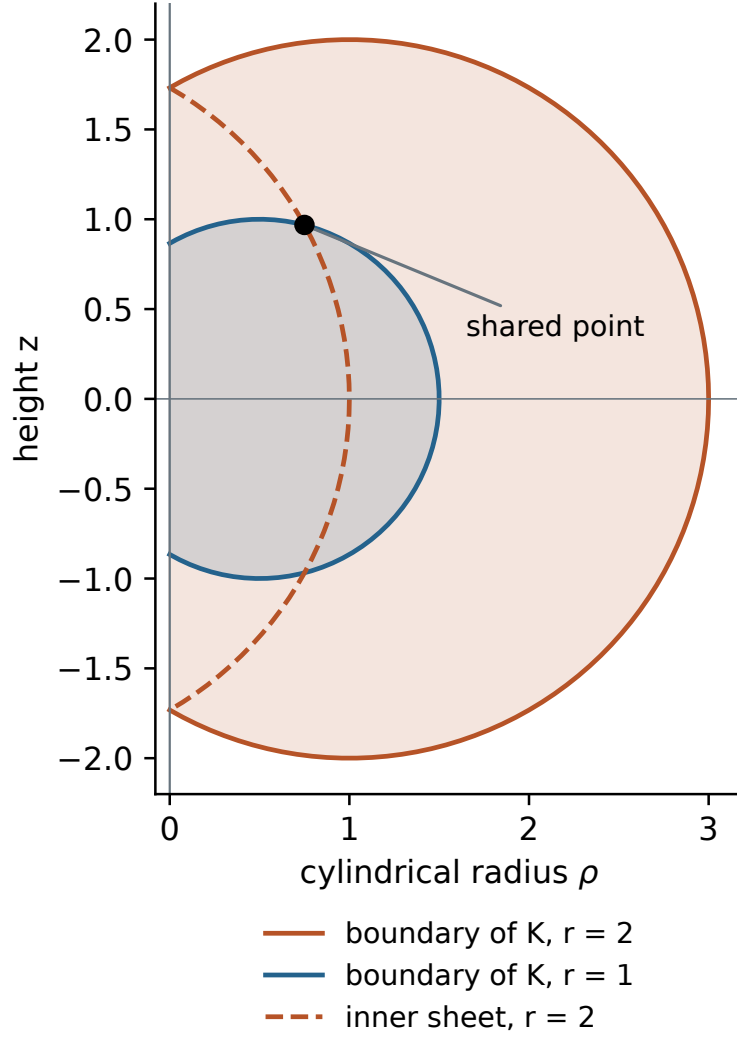


Figure 3: Physical meridian half-plane $\rho \geq 0$ for the exact witness (8). Shaded sections are the diagnostic bodies for $a = 1/2$, $r = 1, 2$. The dashed curve is the larger negative-signed-radius image sheet, folded to $\rho \geq 0$. The marked point lies on that sheet inside the larger body.

4 A prescribed phase family

4.1 Domain and finite harmonic reduction

Fix $\alpha > 0$, a real C , complex amplitudes A_n , and real shell phases b_n . Let $\Phi(t)$ be a prescribed real phase. No differential equation for Φ is supplied. The source choice $b_n = \beta n$ is a special case. Define functions on the common angle domain by

$$\begin{aligned} D_n(\theta, \phi) &= |X_n(\theta, \phi) - Ce_1|^2 \\ &= R_n^2 + r_n^2 + C^2 + 2R_nr_n \cos \theta - 2C(R_n + r_n \cos \theta) \cos \phi, \end{aligned} \quad (9)$$

$$w_n(\theta, \phi) = A_n e^{-\alpha D_n(\theta, \phi)},$$

$$\psi(\theta, \phi; t) = \sum_{n=0}^{N-1} w_n(\theta, \phi) \cos(\Phi(t) + b_n), \quad P = |\psi|^2. \quad (10)$$

Here $e_1 = (1, 0, 0)$. The summands are evaluated at different spatial shell points sharing the same angles. Thus (10) defines a common-domain sum, not yet a spatial superposition at one

common ambient point.

Theorem 4.1 (Three-coefficient phase law). *Define the fixed complex functions*

$$U = \sum_n w_n \cos b_n, \quad V = \sum_n w_n \sin b_n.$$

For a free phase value Φ ,

$$\psi = U \cos \Phi - V \sin \Phi, \quad (11)$$

$$P(\Phi) = c_0 + c_1 \cos(2\Phi) + c_2 \sin(2\Phi), \quad (12)$$

$$c_0 = \frac{1}{2}(|U|^2 + |V|^2), \quad c_1 = \frac{1}{2}(|U|^2 - |V|^2), \quad c_2 = -\operatorname{Re}(U\bar{V}). \quad (13)$$

The real linear span of the phase-indexed functions $P(\Phi)$ has dimension at most three, and P is π -periodic as a function of Φ .

Proof. The addition law for $\cos(\Phi + b_n)$ gives (11). Multiplication by its complex conjugate gives

$$|\psi|^2 = |U|^2 \cos^2 \Phi + |V|^2 \sin^2 \Phi - 2 \operatorname{Re}(U\bar{V}) \sin \Phi \cos \Phi.$$

The double-angle identities yield (12) and (13). Every density in the family is a real linear combination of the three fixed functions c_0, c_1, c_2 ; these need not be independent. The stated phase period follows from the doubled arguments. $\square$

Phase periodicity says nothing by itself about periodicity in t . For example, a phase that scans nonuniformly need not give a periodic time trace. Nor does the rank-three observation supply an evolution equation for amplitudes or radii.

Proposition 4.2 (Modulation and zeros). *Pointwise on $\mathbb{T}^2$,*

$$c_0^2 - c_1^2 - c_2^2 = (\operatorname{Im}(U\bar{V}))^2. \quad (14)$$

Consequently $M = (c_1^2 + c_2^2)^{1/2} \leq c_0$ and the range of P as Φ varies is $[c_0 - M, c_0 + M]$. If U, V are real and not both zero at a point, there is one zero phase modulo π . If both vanish, all phases vanish there.

Proof. Subtracting the first two squared coefficients gives $c_0^2 - c_1^2 = |U|^2|V|^2 = |U\bar{V}|^2$; subtracting c_2^2 leaves the squared imaginary part. Nonnegativity of c_0 proves the bound. The sinusoid $c_1 \cos 2\Phi + c_2 \sin 2\Phi$ has range $[-M, M]$, including the constant case $M = 0$.

For real U, V put $B = \sqrt{U^2 + V^2} > 0$ and choose δ with $(U, V) = B(\cos \delta, \sin \delta)$. Then $P = B^2 \cos^2(\Phi + \delta)$, which has exactly one zero modulo π . The case $B = 0$ is immediate. $\square$

For general complex pairs the density may be strictly positive at every phase. It is constant precisely when $c_1 = c_2 = 0$. Pointwise zeros at different phases must not be mistaken for a simultaneous zero of the whole pattern. A useful simultaneous case is $N = 1$: when $\cos(\Phi + b_0) = 0$, the entire pattern vanishes, for any A_0 . If $A_0 = 0$ it vanishes at every phase. These are exact blink cases, relevant to normalization below.

4.2 Three samples and what they recover

Theorem 4.3 (Known-phase reconstruction). *At each point, let P_0, P_{45}, P_{90} be the intensities at phases $0, \pi/4, \pi/2$, respectively. Then*

$$c_0 = \frac{P_0 + P_{90}}{2}, \quad c_1 = \frac{P_0 - P_{90}}{2}, \quad c_2 = P_{45} - c_0. \quad (15)$$

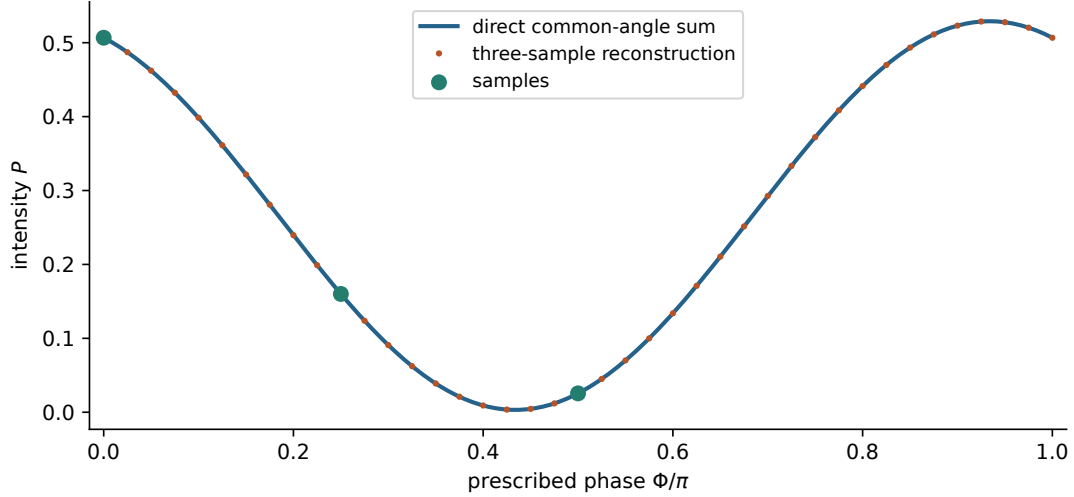


Figure 4: Three-phase recovery at one common-angle point of a ring-family example. Parameters are $N = 3$, $s = 1.5$, $r_0 = 1$, $a = 2$, $C = 1$, $\alpha = 0.08$, $A = (1, 0.7e^{0.6i}, 0.4e^{-0.4i})$, $b = (0, 0.8, 1.7)$ and $(\theta, \phi) = (0.7, 1.1)$. The curve computed from the source sum coincides with the reconstruction from the three marked samples. This is a prescribed phase scan, not a simulated time evolution.

For three arbitrary known phases Φ_a, Φ_b, Φ_c the coefficient matrix with rows $(1, \cos 2\Phi_j, \sin 2\Phi_j)$ has determinant

$$4 \sin(\Phi_b - \Phi_a) \sin(\Phi_c - \Phi_b) \sin(\Phi_c - \Phi_a). \quad (16)$$

Hence unrestricted coefficient triples are uniquely recovered exactly when the known phases are pairwise distinct modulo π .

Proof. At the three specified phases the rows are $(1, 1, 0)$, $(1, 0, 1)$, and $(1, -1, 0)$, giving (15). For the determinant, subtract row a from rows b, c . The difference pair for a phase v relative to u is

$$(\cos 2v - \cos 2u, \sin 2v - \sin 2u) = 2 \sin(v - u)(-\sin(v + u), \cos(v + u)).$$

Factoring the two sine factors from the remaining 2 by 2 determinant leaves $\sin(\Phi_c - \Phi_b)$. This yields (16). The matrix is invertible exactly when all three factors are nonzero. If it is singular, adding any nonzero null vector to a coefficient triple leaves the sampled values unchanged; there is no general linear uniqueness. $\square$

The theorem determines the complete phase-indexed density (12), not the complex amplitude, a phase history, or the individual shell coefficients. It is a known-phase result. Phase-shifting interferometry also recovers quantities from intensities at several shifts [1, Sec. I], but its experimental interpretation and calibration procedures are not assumed here. In particular, our doubled sample phases are $0, \pi/2, \pi$, rather than the equal 120° shifts of the three-bucket procedure discussed by Cheng and Wyant [1, Sec. III.C].

Exact uniqueness also differs from numerical conditioning. As two phases approach congruence modulo π , the matrix approaches a singular matrix. Its inverse sensitivity can therefore become large; the condition number depends on the chosen norm and error model [6]. We neither optimize sample phases nor assert a noise bound. Figure 4 illustrates the exact reconstruction for one declared example.

For the abstract reduction (11), the pairs $(U, V) = (1, i)$ and $(1, -i)$ produce $\psi = e^{-i\Phi}$ and $e^{i\Phi}$, respectively. Both give $P = 1$. This is an amplitude-pair nonuniqueness witness; we do

not claim that these constant pairs are realized by a nonzero restricted Gaussian ring family. A direct ambiguity *within* that family is $A_n \mapsto e^{i\delta} A_n$ for every n , which multiplies ψ by an overall phase and leaves P unchanged. No unique per-shell recovery follows from three density samples.

5 A restricted obstruction to rigid rotation

A varying threshold picture can appear to move without being a rigidly rotated fixed pattern. We now give a precise test for the latter notion. The hypotheses matter: the result below is for the finite family (10), fixed coefficients, ring geometry, and a positive off-axis Gaussian centre.

Theorem 5.1 (No nonzero continuously rotating rigid profile). *Assume $s > 1$, $a > 1$, $C > 0$, $\alpha > 0$, finite $N \geq 1$, and fixed $A_n \in \mathbb{C}$, $b_n \in \mathbb{R}$. Let I be an interval and $\gamma : I \rightarrow \mathbb{R}$ a continuous nonconstant function. Suppose there exists one fixed function H on $\mathbb{T}^2$ such that*

$$P(\theta, \phi; t) = H(\theta, \phi - \gamma(t)) \quad \text{for every } (\theta, \phi) \in \mathbb{T}^2, t \in I. \quad (17)$$

Then P is identically zero on $\mathbb{T}^2 \times I$. In fact, $A_n \cos(\Phi(t) + b_n) = 0$ for every n, t .

Proof. Equation (9) depends on ϕ only through $\cos \phi$. Thus $P(\theta, \phi; t) = P(\theta, -\phi; t)$. Substituting (17), and putting $x = \phi - \gamma(t)$, gives

$$H(\theta, x) = H(\theta, -x - 2\gamma(t)).$$

Composing the reflections for two times shows that H has every angular period $2(\gamma(t) - \gamma(u))$, with either sign immaterial. Since a continuous nonconstant real function on an interval has an image containing a nondegenerate interval, these differences contain a neighborhood of zero. Invariance under all sufficiently small translations implies invariance under every translation: express any real displacement as an integer multiple of a sufficiently small one. Therefore H , and hence each $P(\cdot, \cdot; t)$, is independent of ϕ .

It remains to exclude a nonzero axisymmetric member of this family. Fix $\theta = 0$ and a time t . With $Y = Y(0, \phi)$, the norm $|Y|^2 = (\bar{R} + r_0)^2$ is independent of ϕ . The amplitude takes the form

$$\psi(0, \phi; t) = \sum_{n=0}^{N-1} d_n(t) e^{\lambda_n \cos \phi}, \quad \begin{cases} d_n = A_n \cos(\Phi(t) + b_n) e^{-\alpha(s^{2n}(\bar{R}+r_0)^2 + C^2)}, \\ \lambda_n = 2\alpha C s^n (\bar{R} + r_0). \end{cases} \quad (18)$$

The λ_n are positive and strictly increasing. Put $x = \cos \phi$. Axisymmetry implies that

$$\sum_{n,m} d_n \bar{d}_m e^{(\lambda_n + \lambda_m)x} - K = 0 \quad (-1 \leq x \leq 1) \quad (19)$$

for some real constant $K \geq 0$.

Distinct real exponentials are linearly independent on this interval. Indeed, for distinct exponents $\mu_1, \dots, \mu_\ell$, differentiate a zero combination $0, \dots, \ell - 1$ times at an interior point. After factoring the nonzero $e^{\mu_j x}$ from columns, the matrix is Vandermonde, with determinant $\prod_{i < j} (\mu_j - \mu_i) \neq 0$. All coefficients must vanish.

If any $d_n \neq 0$, let L be the largest such index. Among all terms with nonzero coefficients in (19), the largest exponent is $2\lambda_L$. It arises uniquely from $(n, m) = (L, L)$ and has coefficient $|d_L|^2 > 0$. Mixed sums can coincide with one another, but none reaches this extreme diagonal exponent. The constant term has exponent zero, strictly smaller than $2\lambda_L$. Grouping coincident lower exponents and applying independence gives a contradiction. Thus every $d_n = 0$. The Gaussian factors in (18) are positive, so $A_n \cos(\Phi(t) + b_n) = 0$ for each n . This holds at every t , and (10) then vanishes at every angle. $\square$

The proof is analytic; checking sampled azimuths could not establish its conclusion. It does not require that Φ itself be continuous or nonconstant. A phase held at a complete blink is a possible zero family, even when some $A_n \neq 0$. The conclusion is therefore zero *pattern*, not necessarily zero fixed coefficients.

The theorem does not address a centred choice $C = 0$, a phase depending on azimuth, changing shell coefficients, or an infinite sum. Nor does it exclude nonrigid movement of level sets as the three coefficients are combined with a changing phase. Its role is to separate a specific rigid-rotation assertion from the phase dependence that (10) actually defines.

6 Measures, normalization, and ambient extension

6.1 Area on embedded rings

Assume $a > 1$ throughout this subsection. By (3) the induced first fundamental form and area element are

$$g_n = r_n^2 d\theta^2 + (R_n + r_n \cos \theta)^2 d\phi^2, \quad d\sigma_n = r_n(R_n + r_n \cos \theta) d\theta d\phi. \quad (20)$$

Here σ_n denotes the pullback of Euclidean surface area to the parameter domain. Integrating gives

$$\sigma_n(\mathbb{T}^2) = 4\pi^2 R_n r_n. \quad (21)$$

These are the classical ring-torus metric and area formulas [15, Eqs. 22–24, 28–29], with our angles ordered as meridian then azimuth.

The coordinate measure $\nu = d\theta d\phi$ and the area measure are mutually absolutely continuous, because

$$0 < r_n(R_n - r_n) \leq \frac{d\sigma_n}{d\nu} \leq r_n(R_n + r_n) < \infty.$$

They are not identical, nor constant multiples of one another: the factor depends nontrivially on θ . Thus a density with respect to coordinate measure is not automatically a density with the same values with respect to surface area.

Proposition 6.1 (Normalization and homothetic area measures). *For a fixed phase, let P be the common-domain observation in (10). Define $Z_n = \int_{\mathbb{T}^2} P d\sigma_n$. Then*

$$d\sigma_n = s^{2n} d\sigma_0, \quad Z_n = s^{2n} Z_0.$$

If $Z_0 > 0$, the densities $p_n = P/Z_n$ satisfy

$$p_n = s^{-2n} p_0, \quad p_n d\sigma_n = p_0 d\sigma_0. \quad (22)$$

All these integrals are finite. A complete zero pattern has $Z_n = 0$ and admits no normalization by this rule.

Proof. Each factor in the area weight in (20) scales by s^n , proving the measure relation. Integration gives the relation for Z_n , and division when $Z_0 > 0$ gives (22). The finite sum (10) is continuous on the compact domain, so P is bounded and each integral is finite. With a strictly positive continuous area weight, a continuous nonnegative P has zero integral if and only if it vanishes everywhere: a positive value would give a neighborhood of positive integral. Division by zero is not a normalized density. $\square$

Equality in (22) is equality of measures pulled back to the *common angle domain*. Their spatial pushforwards use different maps X_n . For a nonzero member of this analytic family, P cannot vanish on an open parameter set, so these pushforwards have supports S_n , which are different spatial sets. One way to see the analytic assertion is to restrict a convergent local real power series to lines: vanishing on an open set forces all its coefficients there to zero, and continuation across overlapping charts on the connected torus propagates that identity. Nonzero P is therefore positive somewhere in every open set. The different supports also follow geometrically from the distinct maximum cylindrical radii $R_n + r_n$. Spatial shell measures are not being equated, even if some images intersect.

More generally choose any fixed finite positive measure μ for which U, V are square-integrable. Integrating the phase identity gives

$$\begin{aligned} Z(\Phi) = \frac{1}{2}(\|U\|_\mu^2 + \|V\|_\mu^2) + \frac{1}{2}(\|U\|_\mu^2 - \|V\|_\mu^2) \cos 2\Phi \\ - \operatorname{Re} \left(\int U \bar{V} d\mu \right) \sin 2\Phi. \end{aligned} \quad (23)$$

It is constant for *all phase values* exactly when $\|U\|_\mu^2 = \|V\|_\mu^2$ and $\operatorname{Re} \int U \bar{V} d\mu = 0$. Necessity follows from linear independence of $1, \cos 2\Phi, \sin 2\Phi$; sufficiency is immediate. A constant time trace alone need not imply these conditions if the prescribed phase stays fixed. If its range contains a nondegenerate interval, the same independence does apply.

No historical measure or display threshold is supplied by these choices. Calling P a density is harmless as shorthand for a nonnegative observation only if its measure-dependent normalization has not silently been assumed.

6.2 An extension that can be written, but is not unique

Let $a > 1$ and retain the reference chart Y . Homothety gives

$$|s^n Y - C e_1|^2 = s^{2n} |Y - C s^{-n} e_1|^2.$$

It follows that the explicitly defined ambient function

$$F_t(x) = \sum_n A_n \exp\left(-\alpha s^{2n} |x - C s^{-n} e_1|^2\right) \cos(\Phi(t) + b_n), \quad x \in \mathbb{R}^3, \quad (24)$$

satisfies $\psi = F_t \circ Y$. This is an exact reference-shell pullback representation. It relocates the centres to $C s^{-n} e_1$ and changes the precisions to αs^{2n} . It is not the assertion that the original terms already sampled one ambient point.

Proposition 6.2 (Extension freedom at a fixed shell). *Writing ambient Cartesian coordinates as (x, y, z) , define*

$$Q(x, y, z) = (x^2 + y^2 + z^2 + \bar{R}^2 - r_0^2)^2 - 4\bar{R}^2(x^2 + y^2). \quad (25)$$

For any smooth complex function h_t on $\mathbb{R}^3$, $F_t + Q h_t$ has the same pullback to Y as F_t . There exist choices that differ from F_t away from S_0 .

Proof. On Y , put $q = \bar{R} + r_0 \cos \theta > 0$. Then $x^2 + y^2 = q^2$, $z = r_0 \sin \theta$, and $x^2 + y^2 + z^2 + \bar{R}^2 - r_0^2 = 2\bar{R}q$. Substitution gives $Q \circ Y = 0$. Hence $(F_t + Q h_t) \circ Y = F_t \circ Y = \psi$. For example, $h_t = 1$ changes the value at the origin because $Q(0, 0, 0) = (\bar{R}^2 - r_0^2)^2 > 0$ in the ring case. $\square$

The freedom exists even when the reference shell is fixed. It proves that the surface data do not determine a unique ambient extension. It does not prove that ambient fields are impossible, choose a physical one, or recover a field equation. Neither (24) nor (25) introduces coupling to a separate native computational shell.

7 Display maps and retained information

The maps in this section have different inputs from (10). They are analyzed as maps of records. Their parameters are display conventions, and their coordinates do not establish a connection to the carrier family. All inverses here are exact mathematical statements, not claims that a numerical inverse exists in the source software.

7.1 A whole-history torus display

Use $M \geq 1$ for an integer clock size, to distinguish it from the number of shells. The source parameter for this clock is named N , with historical default 12. Given a nonempty finite history (κ_j, q_j, z_j) , require finite real $\kappa_j \geq 0$, z_j and integer q_j . Let $R > r_{\max} > 0$ and set

$$\begin{aligned} H_z &= \max_j |z_j| + \varepsilon, \quad \varepsilon = 10^{-9}, & \vartheta_j &= \frac{2\pi(q_j \bmod M)}{M}, \\ r_j &= r_{\max} \frac{\kappa_j}{1 + \kappa_j}, & \chi_j &= \frac{\pi z_j}{2H_z}. \end{aligned} \quad (26)$$

The literal regularizer is part of the numeric display convention; its addition presupposes the same numeric units as z_j . In exact arithmetic, $0 \leq r_j < r_{\max}$ and $|\chi_j| < \pi/2$. The displayed point is

$$(X_j, Y_j, Z_j) = ((R + r_j \cos \chi_j) \cos \vartheta_j, (R + r_j \cos \chi_j) \sin \vartheta_j, r_j \sin \chi_j). \quad (27)$$

The scalar input z_j and displayed height Z_j are different quantities.

Theorem 7.1 (History coordinates and exceptional fibre). *For known $R, r_{\max}, H_z, M$ and a point in the image of (27) with $\kappa_j > 0$, define $\rho_j = \sqrt{X_j^2 + Y_j^2}$. Then*

$$\begin{aligned} r_j &= \sqrt{(\rho_j - R)^2 + Z_j^2}, & \kappa_j &= \frac{r_j}{r_{\max} - r_j}, \\ \chi_j &= \text{atan2}(Z_j, \rho_j - R), & z_j &= \frac{2H_z \chi_j}{\pi}, \\ \vartheta_j &= \text{atan2}(Y_j, X_j) \pmod{2\pi}. \end{aligned} \quad (28)$$

The last angle recovers $q_j \bmod M$, not its unbounded integer representative. At $\kappa_j = 0$, XYZ retains this residue but discards z_j . A record that also retains χ_j and H_z still recovers z_j at zero.

Proof. The adopted domain gives $R + r_j \cos \chi_j > 0$ and, when $r_j > 0$, $\rho_j - R = r_j \cos \chi_j > 0$. Thus the right-half-plane branch of atan2 uniquely recovers $\chi_j \in (-\pi/2, \pi/2)$. The radial norm recovers r_j , and the strictly increasing function $\kappa \mapsto r_{\max} \kappa / (1 + \kappa)$ has the stated inverse on $[0, r_{\max})$. The azimuth is defined because $\rho_j > 0$, and its discrete value determines the residue.

If $\kappa_j = 0$, the point is $(R \cos \vartheta_j, R \sin \vartheta_j, 0)$ independently of z_j , so no scalar inverse from XYZ alone exists. This failure can occur even with fixed known H_z : include another history row with a fixed larger absolute scalar, and vary the zero-radius row's scalar below that maximum. The richer record directly gives $z_j = 2H_z \chi_j / \pi$, so its fibre is different. $\square$

The native `HistoryTorus` record explicitly retains r, χ, H_z , the maximum, regularizer, radii and clock size [13]. Loss from XYZ must not be attributed to that richer record. Conversely, if H_z is unavailable, (28) supplies only z_j / H_z , not an absolute scalar inverse for an arbitrary point.

The display consists of finitely many points in discrete azimuthal sectors, with varying tube radii. It is not generally one two-dimensional torus surface. Appending a row with a larger $|z|$

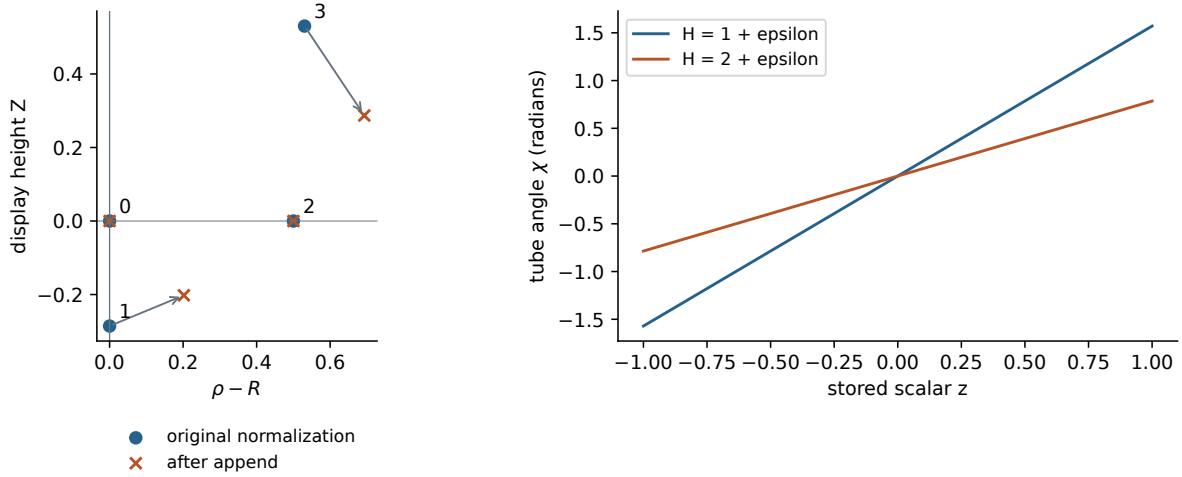


Figure 5: A history display in tube-plane coordinates $(\rho - R, Z)$, which remove only the known azimuth. With $R = 2, r_{\max} = 1, M = 12$, histories share $\kappa = (0, 0.4, 1, 3)$, $q = (0, 1, 4, 7)$, $z = (0.6, -1, 0, 0.5)$. Adding a row $(1, 2, 2)$ changes the normalization from $1 + 10^{-9}$ to $2 + 10^{-9}$. Arrows connect the earlier points before and after this change. The zero-radius row and the zero-scalar row stay fixed. The right panel displays the two scalar-to-angle rules.

changes H_z and therefore generally changes earlier displayed points. Exceptions include earlier rows with $z_j = 0$ or $\kappa_j = 0$; if the maximum does not change, no prior point moves for this reason. The stored history values themselves remain unchanged.

7.2 Cylinder and direct vectors

For the same scalar and clock inputs the cylinder display is

$$(X, Y, Z) = (\kappa \cos \vartheta, \kappa \sin \vartheta, z). \quad (29)$$

For $\kappa > 0$ its inverse gives $\kappa = \sqrt{X^2 + Y^2}$, $\vartheta = \text{atan2}(Y, X)$ modulo 2π , and $z = Z$. At $\kappa = 0$ it retains the scalar height, but every admissible clock residue maps to $(0, 0, z)$. The azimuthal loss is nontrivial when $M > 1$; with $M = 1$ there was only one residue. This is the opposite zero-radius scalar behavior from (27).

Direct-vector extraction copies the three components of a selected recorded vector. In the native source the default key is `Z_total`, with `Z_vec` used only when that default is absent [13]. The map is an identity on the selected vectors in their existing frame. This fact alone does not make it injective on an unspecified complete state. One would need a definition of that state and of its observation map to ask the larger inversion question.

7.3 A labelled logarithmic channel map

The historical three-channel viewer uses fixed anchors $\phi_j = 0, 2\pi/3, 4\pi/3$ [12]. Analyze one retained label j in its known Cartesian frame. For $R > r_0 > 0$, $c > 0$, and a nonzero complex input Ω , set

$$\rho_c = r_0[1 + c \log(1 + |\Omega|)], \quad \theta = \arg \Omega \pmod{2\pi}, \quad (30)$$

and

$$(X, Y, Z) = ((R + \rho_c \cos \theta) \cos \phi_j, (R + \rho_c \cos \theta) \sin \phi_j, \rho_c \sin \theta). \quad (31)$$

The subscript on ρ_c distinguishes this tube radius from the nonnegative spatial cylindrical radius. Its logarithmic growth is unbounded. The original numeric formula implicitly uses a dimensionless or already scaled modulus inside $\log(1 + |\Omega|)$.

Proposition 7.2 (Labelled inverse and axis crossing). *For a point in the admitted image, retain the signed anchor projection*

$$u = X \cos \phi_j + Y \sin \phi_j.$$

Then

$$\begin{aligned} \rho_c &= \sqrt{(u - R)^2 + Z^2}, & m &= \exp\left(\frac{\rho_c/r_0 - 1}{c}\right) - 1, \\ \Omega &= m \frac{u - R + iZ}{\rho_c}. \end{aligned} \tag{32}$$

For nonzero input this is an exact inverse. A zero input is recovered as zero from its conventionally chosen image; no intrinsic argument for zero is required. Negative signed projection occurs exactly when $\rho_c > R$ and $\cos \theta < -R/\rho_c$.

Proof. Since the anchor vector is a unit vector, (31) gives $u = R + \rho_c \cos \theta$, including when $u < 0$. The norm of $(u - R, Z)$ recovers $\rho_c \geq r_0 > 0$. Inverting the logarithm gives $m = |\Omega|$, and division by ρ_c recovers $\cos \theta + i \sin \theta$. Their product is Ω . At zero, $m = 0$, so that product recovers zero for any explicitly chosen forward angle convention. The historical numeric routine uses its library's zero-angle convention; the complex number zero itself has no argument. Finally $u < 0$ is exactly $R + \rho_c \cos \theta < 0$, giving the stated pair of conditions. $\square$

The label matters: replacing u by $\sqrt{X^2 + Y^2} = |u|$ folds the negative branch and invalidates this inverse. At the shared axis different labels also lose their distinct anchor directions. Without the label, the union of channels need not be injective. With the label retained, the signed inverse remains valid at an axis point.

Large modulus *permits* crossing when $|\Omega| > \exp((R/r_0 - 1)/c) - 1$, but does not force it; phase $\theta = 0$ always gives positive u . The visual background wireframe has fixed tube radius r_0 . It is not the surface carrying all the channel points.

Proposition 7.3 (Nonzero re-entry into the background wireframe). *Let the background ring be*

$$W = \{(X, Y, Z) : (\sqrt{X^2 + Y^2} - R)^2 + Z^2 = r_0^2\}.$$

On the branch $u \geq 0$, a channel point lies on W only for $|\Omega| = 0$. On the branch $u < 0$, it lies on W precisely when

$$2R - r_0 \leq \rho_c \leq 2R + r_0, \quad \cos \theta = \frac{r_0^2 - 4R^2 - \rho_c^2}{4R\rho_c}. \tag{33}$$

Thus the admissible nonzero modulus band is

$$\exp\left(\frac{2R/r_0 - 2}{c}\right) - 1 \leq |\Omega| \leq \exp\left(\frac{2R/r_0}{c}\right) - 1, \tag{34}$$

with the phase condition in (33) still necessary.

Proof. The cylindrical radius of a channel point is $|u|$. If $u \geq 0$, its squared distance from the background meridian centre is $(u - R)^2 + Z^2 = \rho_c^2$, which equals r_0^2 only for $\rho_c = r_0$, hence $|\Omega| = 0$.

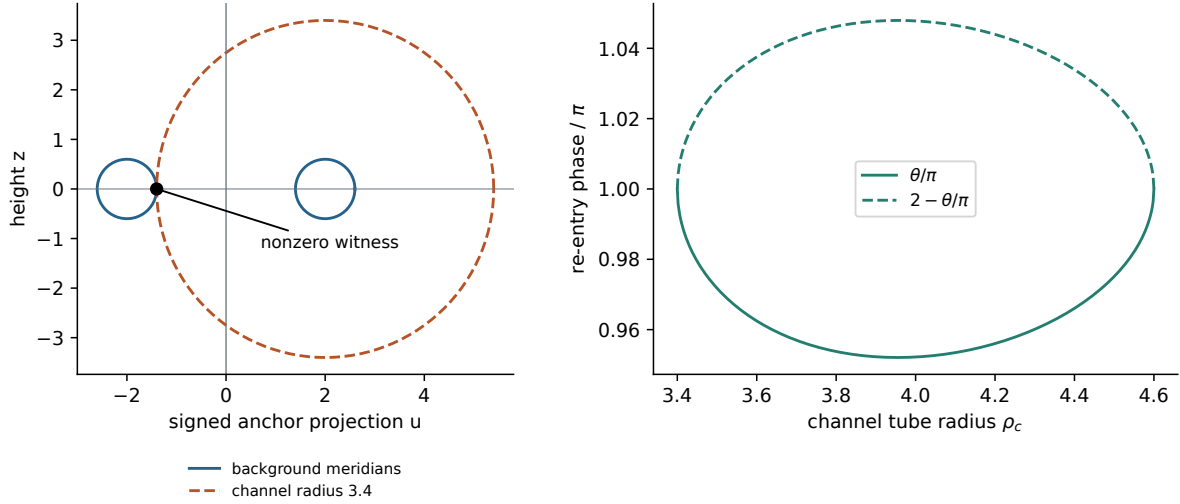


Figure 6: Labelled channel geometry at $R = 2, r_0 = 0.6, c = 0.4, \phi_j = 0$. Left: the signed anchor plane (u, Z) shows the two background meridian circles and the tube-radius $\rho_c = 3.4$ locus (dashed); the nonzero witness is $(-1.4, 0)$. Right: the allowed negative-branch re-entry phases over $3.4 \leq \rho_c \leq 4.6$. A radius in this band alone does not put a point on the wireframe.

If $u < 0$, using $|u| = -u$ gives

$$(|u| - R)^2 + Z^2 = \rho_c^2 + 4R\rho_c \cos \theta + 4R^2.$$

Equating this to r_0^2 gives the cosine in (33). That cosine is at most 1 automatically, because $r_0 < R$ and its numerator is negative. It is at least -1 exactly when $(\rho_c - 2R)^2 \leq r_0^2$, proving the radius band. On that band $\rho_c \geq 2R - r_0 > r_0$, and substitution yields $u = (r_0^2 - \rho_c^2)/(4R) < 0$, so the branch condition is indeed satisfied. Monotonic inversion of (30) gives (34). $\square$

At the historical defaults $R = 2, r_0 = 3/5, c = 2/5$, choose

$$\Omega = -(\exp(35/3) - 1), \quad \phi_j = 0. \quad (35)$$

Then $\theta = \pi$, $\rho_c = 17/5$, and the point is $(-7/5, 0, 0)$, which lies on W since $(7/5 - 2)^2 = 9/25 = r_0^2$. This nonzero witness prevents the incorrect assertion that only a zero channel can ever meet the background. The upper endpoint of the modulus band is $\exp(50/3) - 1$. These are properties of the display map; they do not establish that a native trajectory reaches these inputs.

7.4 Exact inverses and finite precision

Exact branch and side-information conditions are part of each theorem. Floating arithmetic can round $\kappa/(1 + \kappa)$ to one, round H_z to the maximum despite the positive regularizer, or lose accuracy in $\rho - R$. A successful finite floating result therefore does not certify the strict inequalities used by Theorem 7.1. The native routine's documentation explicitly warns about such saturation [13]. The channel exponential can also overflow for large decoded radii. These cautions limit numerical use of the formulas; they do not alter the exact fibres proved above. We have not implemented or certified a runtime inverse.

Colour and percentile displays introduce additional normalization or clipping, summarized in Appendix B. They require their own retained scale and frame information. A coordinate loss in one map must not be generalized to all stored records or to a different display.

8 Consequences and limits

The full-image intersection interval, strict spindle-body containment, and the exact shared witness answer different geometric questions. A homothetic chart family can have intersecting spindle images while its diagnostic bodies are strictly nested. The positive result is a precise classification, rather than a single ambiguous claim about overlap.

The prescribed common-angle sum yields an equally concrete observation law: three real coefficient functions determine its entire phase-indexed squared modulus. Three suitable known samples recover these coefficients; complex amplitudes and individual shell weights require other information. The restricted rigid-rotation theorem then uses the actual azimuthal dependence, not the appearance of selected plots.

Measures and displays complete the information accounting. Normalization is conditional on a chosen measure and a positive integral. An ambient extension can be constructed, but is not uniquely determined by its restriction. A history-torus XYZ record, its richer retained record, a cylinder, and a labelled logarithmic channel have different exceptional fibres. These distinctions are sufficient to describe the defined mathematics without assigning any of the objects a physical field interpretation.

A Bounded source correspondence

The primary equations for the radii, chart, Gaussian weights, phase sum and squared modulus occur in *Converging Symmetries*, version dated 27 May 2025, physical pages 1–2, Eqs. 1–8 [9]. The source’s Gaussian-centre notation is unindexed R ; we use C to avoid identifying it with the unscaled radius parameter R_* . The historical source denotes that parameter by R_0 ; here R_* distinguishes it from the indexed zeroth radius $R_{n=0} = \bar{R}$. This notation change preserves the radii. A later version dated 28 May 2025 has a different printed author line [8]. The title, date and supplied edition are retained in the source ledger; it is not silently treated as the same bibliographic item. The later recursive-field paper reproduces the scaled construction on physical page 17, Sec. 19.1, Eqs. 18–19 [7]. Its presentation does not by itself supply a map from the common-angle scalar observation to a spatial field with an evolution equation.

The visually confirmed physical page 4 of that paper contains two further, distinct prescriptions [3]. In Sec. 3.1 the curve family is

$$x = (R + 0.1i) \cos t, \quad y = (R + 0.1i) \sin t, \quad z = 0.2(1 - 0.15i) \sin 2t. \quad (36)$$

Here i is a curve index, not the imaginary unit i ; its range is not specified in that prescription. Section 3.2 gives code for five curves $i = 0, \dots, 4$, with 200 equally spaced t values from 0 to 2π and

$$\begin{aligned} \delta &= 0.1 \sin(5t + i), \\ x &= (R + r \cos(t + i) + \delta) \cos t, \\ y &= (R + r \cos(t + i) + \delta) \sin t, \quad z = r \sin(t + i). \end{aligned} \quad (37)$$

These are prescribed curves, not the sum (10). For the continuous expression (37), take $i = 0, t = \pi/2$, with $R, r > 0$. It gives $(0, R + 0.1, r)$, whose squared meridian distance from the unperturbed torus core is $r^2 + 0.01$, not r^2 . This exact witness concerns the formula; it need not be one of the 200 sampled points.

The historical file `recursive_field_evolve.py` does contain an actual coordinate-update algorithm [10]. Its `evolve_field` constructs five-nearest-neighbor adjacency, uses a mean neighbor displacement and its componentwise hyperbolic tangent, adds a centre-directed bias

and random perturbation, and applies a step-dependent multiplier to the third coordinate. The adjacency is constructed before the step loop. Some named constants and the computed breathing amplitude do not enter the coordinate update. These facts follow from text inspection, not execution. No explicit source map identifies this algorithm with (10), its extension, or a native computational shell.

The native and historical coordinate adapters are separate sources for Section 7 [13, 11, 12]. A historical fixed 12-sector angle and the native modular clock agree for integer clock inputs at that clock size. Their source lineage does not turn the Gaussian carrier family into the input state of the display.

The remaining historical gaps are narrow: definitions and domains of the recursive operators denoted G, T ; a specified norm or metric and field type for related operator claims; a measure and numerical threshold for the historical intensity display; and a transport or consumer map joining the distinct constructions. A time-only seed does not exclude unknown operators introducing spatial dependence. Likewise a scalar expression does not supply a vector curl without a declared vector-valued reading. These missing definitions are recorded, not completed by conjecture. The page-4 confirmation and received final addendum are closed evidence items, not current gaps.

B Colour and percentile transformations

The viewer's trail colour uses $(J - J_{\min})/(J_{\max} - J_{\min})$ when the range is positive, and zero for a constant series [12, lines 419–424]. Even before a colour map, the normalized values need the extrema to recover the original scale. The constant branch collapses every constant level to zero unless that level is retained. A displayed RGB value additionally depends on the colour map and its possible quantization; it is not assumed to be an exact inverse code for a scalar.

A different spectral display floors energies at 10^{-12} , takes their logarithms and then applies range normalization; it uses a separate clipped brightness factor [12, lines 537–549]. All values below the floor have the same transformed energy. Retaining extrema can invert a nonconstant logarithmic range above the floor, but cannot undo the floor itself. Retaining a coordinate array also does not undo a clipped colour channel.

The vector overlay uses the 99th percentile d of finite vector norms greater than 10^{-12} , and multiplies the selected vectors by $0.85r_{\text{frame}}/d$ [12, lines 652–665]. When that selection is empty, or the resulting denominator is nonfinite or at most 10^{-12} , the code uses $d = 1$. Recovering an input vector from a scaled plotted vector requires the actual scale and frame; selecting one vector array is still not a definition of a complete state. These transformations are reported individually rather than combined into a global claim that all display coordinates lose the same information.

C Reproducibility, attribution, and review status

The accompanying supplement contains the editable source, bibliography, six reproducible figures and their declared numeric data, one isolated checker, exact command records, compiler logs, and a page-by-page visual receipt. The generators are collected in one script. Checks transcribe the mathematical definitions independently; they do not import or execute the project runtime or replay earlier research suites. Exact symbolic identities, exact witnesses, and finite numerical illustrations are distinguished in the results. The analytic proofs of theorems, particularly Theorem 5.1, do not depend on a finite test count.

The proofs apply classical tools to the stated family. The body K_n , the extension fam-

ily, and the selected figure parameters are declared mathematical constructions. The spindle and wireframe examples are counterexamples to broader identifications, not historical measurements. The accompanying proof/source/literature inventory records these roles. Literature checking is targeted, with no claim of comprehensive novelty clearance.

Hilmir Frímánn Halldórsson is the sole named author of this manuscript. Codex prepared the draft and this revision, its derivations, figures, checks and typesetting under the owner's work order. GPT supplied scientific reconciliation and accepted qualifications. Claude contributed an independent prior research pass and the received reconciliation addendum; that prior work is not a review of this manuscript. Some historical sources explicitly credit GPT or ChatGPT in their printed author lines, retained in the bibliography.

Version 0.1 received a GPT manuscript review and a Codex author re-review. Codex also drafted that edition; its re-review is not an independent Claude manuscript review, regardless of the retained legacy report filename. Claude's earlier independent research pass and reconciliation addendum remain distinct from manuscript review. No external peer review or journal acceptance is claimed.

This v0.1.1 repository-research candidate applies the accepted radius-notation and phase-wording corrections and updates this review account. Final GPT verification of these revised bytes and confirmation of the exact public payload remain pending. The separate lens v0.1.1 edition retains its accepted bytes. No publication or release has occurred in this task, and no new licence is selected.

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